

EDUCATION

University of Toronto

B.A.Sc. in Computer Engineering + PEY

- *Minor in Artificial Intelligence; Certificate in Cybersecurity*

Toronto, Canada

Sept 2022 – Jun 2027

cGPA: 3.64/4.0 – Dean's List

WORK EXPERIENCE

Qualcomm, formerly Alphawave SEMI

Physical Design Engineer Intern

Toronto, Canada

May 2025 – Present

- Automated and standardized a **Fusion Compiler** physical design flow, introducing guardrails, regression and trial tooling, and user-safe interfaces to improve run reliability and scalability across teams.
- Integrated **PrimeTime** for **DMSA timing analysis** and **ECO fixing**, enabling robust multi-corner closure and improved post-route timing convergence.
- Built scalable QoR infrastructure and **LLM-powered RAG agents** to collect, analyze, and query timing, power, utilization, and DRC metrics, enabling natural-language QoR review, trend analysis, and actionable next-step recommendations.
- Designed and deployed a cross-team design handoff system to manage flow collateral, ensuring version consistency, traceability, and reproducibility across the ASIC development lifecycle.

Bank of Montreal (BMO)

Junior Software Developer

Toronto, Canada

May 2024 – Aug 2024

- Developed and implemented an **online self-serve platform** using **Angular** and **Node.js**, allowing users to deploy and manage **Kafka topics, access controls, and data schemas** without relying on CLI-based workflows.
- Created and optimized **Python scripts** to proactively monitor **Apache Kafka connectors**, automating **real-time alerts** and enhancing operational efficiency across the team.
- Played a pivotal role in **migrating Kafka streams from EC2 to ECS**, leading to improved **system performance** and significant **cost savings**.

Expert Geophysics Limited (EGL)

Engineering Intern

Aurora, Canada

May 2023 – Aug 2023

- Developed and tested **C-based software** for **airborne magnetic field monitoring systems**, ensuring optimal performance through rigorous design and testing in **critical operations**.
- Conducted testing on key components including the **console, current generator, and power supply**, ensuring **system reliability and functionality**.
- Analyzed and improved the **structural integrity of magnetometer mounts**, implementing **3D-printed designs** that increased the **safety factor by 33%** and reduced **maximum displacement by 83%**.

PROJECTS

BMO Innov8 Hackathon

Angular, Node.js, Apache Kafka, Figma

Toronto, Canada

- **Won 1st place** at the **BMO Innov8 Hackathon**, a competitive internal innovation event with **over 300 participants** and **30 teams**.
- Developed an innovative financial technology solution showcasing advanced problem-solving, product design, and technical execution.

SKILLS

Certifications: Amazon Web Services Certified Cloud Practitioner, Oct 2021 – Sept 2027

Programming Languages: Python, C/C++, Java, JS/TS, SQL, TCL, Bash, Perl, MATLAB, Assembly

EDA & Hardware Tools: Fusion Compiler, PrimeTime, StarRC, IC Validator, Formality, Verilog

Technologies: Linux, Git, SVN, SLURM, Streamlit, LangChain, Ollama, Docker, Apache Kafka, AWS, Google Cloud Platform, NumPy, Pandas, Matplotlib, YAML, JSON, Makefiles

Competencies: Physical Design Methodology, Place and Route, Static Timing Analysis, Timing Closure, QoR Analysis, Scripting and Automation